

Lab: Neural Circuits for Planning and Prospection

Learning Goal: Map planning functions to neural circuits and analyze the balance between goals and habits.

Part 1: DMN Activation Analysis

Exercise: For each mental activity, predict whether the DMN is active or suppressed, and explain why:

Activity	DMN Active or Suppressed?	Explanation
Solving a math problem		
Daydreaming about vacation		
Reading a novel and imagining the characters		
Playing an intense video game		
Planning what to have for dinner		

Write your response here

Part 2: Memory and Imagination

Learning Goal: Explore the shared neural basis of remembering and imagining.

Exercise: Pick a familiar location (your home, school, park).

1. Recall a specific memory from that location — describe it vividly.
2. Now imagine something that has *never happened* at that location — describe it vividly.
3. Notice: Both exercises recruited the same mental machinery. How were they similar?
How did they differ?
4. A patient with hippocampal damage can't do either. Why does this support the idea that memory and imagination share neural circuits?

Write your response here

Part 3: Goals vs. Habits

Learning Goal: Analyze the competition between goal-directed and habitual systems.

Scenarios: For each, identify whether behavior is primarily goal-directed or habitual, and which neural system is dominant:

1. Your usual morning routine (shower, coffee, commute)
2. Navigating a detour because your usual route is blocked
3. Checking your phone every few minutes without a specific purpose
4. Carefully choosing a birthday gift for a close friend

For scenario 3: How might Active Inference explain the automatic phone-checking habit?

Write your response here

Part 4: Constructing a Future Scenario

Learning Goal: Practice hippocampal constructive memory.

Exercise: Imagine yourself five years from now in a specific scenario. Write a detailed description covering:

1. Where you are (spatial details)
2. What you're doing (action details)
3. Who you're with (social details)
4. How you feel (emotional details)

Now analyze: Which of these details came from existing memories? How did your hippocampus recombine them into a novel scenario?

Write your response here

Part 5: Reflection

In 150 words, reflect: The same brain network supports remembering the past, imagining the future, and taking another person's perspective (Theory of Mind). What does this convergence tell us about the fundamental purpose of memory? Is memory primarily for the past or for the future?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Network identification	DMN activation patterns
2	Phenomenological analysis	Shared memory-imagination circuits
3	System classification	Goal-directed vs. habitual control
4	Constructive imagination	Hippocampal scene construction
5	Integrative reflection	Memory's purpose — past or future?